Emergent distance is force-blind: why a correlator-built geometry cannot carry the holographic force

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RETRACTION NOTICE (2026-08-17): the force exponent $\alpha = 3.92 \pm 0.16$ stated in this note is an estimator artifact (95.8% of the fit objective sits on one point). The measured value is 2.6 ± 0.2 and the channel is open. Interpretations built on $\alpha = 4$ are withdrawn. See `research/ERRATA.md` and the Zenodo v5 correction, doi:10.5281/zenodo.22080720.

Abstract

In a critical-Ising multiscale entanglement renormalization ansatz (MERA), a two-point correlator $C(r)$ can be turned into an emergent distance via the Synge world-function map $d(r) = \sqrt{-2\,\epsilon^2 \ln C(r)}$ of Ansel [1]. We show that this distance, while a legitimate and monotone emergent-geometry object, is *force-blind*: its logarithmic slope is set by the lightest primary that appears in $C(r)$ — the spin field σ with scaling dimension $\Delta_\sigma = 1/8$ — giving $\kappa = 2\Delta_\sigma = 1/4$, whereas the holographic force measured elsewhere in this program scales with exponent $\alpha \approx 4$ and is carried by the stress tensor T with $\Delta_T = 2$. We confirm this with a three-tier pre-registered test. On a synthetic CFT correlator $C = r^{-1/4}$ the map returns $\kappa = 0.250$ exactly; on the exact 16-qubit critical transverse-field Ising ground-state connected $\langle Z_i Z_j \rangle$ we measure $\kappa_{\mathrm{chord}} = 0.243$; a known-pathological control fails the monotonicity gate as predicted. The result is **SUPPORTED**. Because distance and force are carried by operators of different dimension, a geometry reconstructed from a single correlator *cannot*, even in principle, reproduce the stress-tensor force. This closes the "correlator → geometry → force" thread and explains, with one clean measurement, why two earlier and structurally distinct approaches in this program failed.

1 Setup: the MERA holographic dictionary and the $\alpha \approx 4$ force

The multiscale entanglement renormalization ansatz (MERA) is widely read as a discrete realization of the holographic principle: its layered causal structure reproduces a hyperbolic (AdS-like) bulk emerging from a critical boundary conformal field theory [2, 3]. Within this dictionary one can ask whether the bulk supports a genuine *force* between boundary insertions. The companion paper to this note answers that affirmatively for the critical Ising chain: a holographic force $F(\Delta r) \sim \Delta r^{-\alpha}$ is observed with

$$\alpha \approx 4, \tag{1}$$

established by three independent measurements that converge on the same value — $\alpha = 3.92$ (24-qubit exact diagonalization), $\alpha = 3.97$ (16-qubit exact), and $\alpha = 4.07$ (16-qubit $\chi = 3$ MERA).[1]

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[1] Companion preprint, Zenodo record 10.5281/zenodo.19661227 (v3).

The exponent $\alpha \approx 4$ is the signature of *stress-tensor exchange*: in a two-dimensional CFT the stress tensor T has scaling dimension $\Delta_T = 2$, and a T-mediated interaction between two boundary operators falls as $r^{-2\Delta_T} = r^{-4}$. This force sits in the same lineage as the "hologron" excitations of Sahay, Lukin and Cotler [4] and the boundary-CFT primary modes of Wang and He [5], which together frame what kinds of bulk excitations a critical-chain tensor network can carry.

The question taken up here is narrower and prior: before asking whether the bulk has a force, can one even reconstruct a bulk *distance* from boundary data, and if so, what does that distance know about? A natural and tempting route is to build the geometry from a two-point correlator. We show below that this route yields a real distance that is nonetheless deaf to the stress tensor, and therefore deaf to the force of Eq. (1).

2 The Synge map and the operator-dimension argument

Ansel [1] proposes that an emergent distance can be read off a two-point function through the world-function (Synge) map

$$d(r) \;=\; \sqrt{-2\,\epsilon^2\,\ln C(r)}, \tag{2}$$

where $C(r)$ is a (normalized, $0 < C \leq 1$) correlator between sites separated by r, and ϵ is a fixed length scale. The construction is geometrically sensible: a perfectly correlated pair ($C \to 1$) sits at zero distance, decorrelation drives d upward, and for a monotonically decaying $C(r)$ the resulting $d(r)$ is monotone increasing. Equation (2) is the object we test.

What sets the slope. At a 2D critical point a two-point function of a primary $\mathcal{O}$ of scaling dimension $\Delta_{\mathcal{O}}$ decays as a power law,

$$C(r) \;\sim\; r^{-2\Delta_{\mathcal{O}}}. \tag{3}$$

We define the dimensionless log-slope of the correlator

$$\kappa \;\equiv\; -\,\frac{d\ln|C|}{d\ln r} \;=\; 2\Delta_{\mathcal{O}}, \tag{4}$$

so that κ directly reports *which operator dominates* the correlator at the separations probed. This is the crux. A correlator built from the spin field σ — the connected $\langle Z_i Z_j\rangle$ of the Ising chain — is dominated by the lightest primary, $\Delta_\sigma = 1/8$, so

$$\kappa_\sigma \;=\; 2\Delta_\sigma \;=\; \tfrac{1}{4} \;=\; 0.25. \tag{5}$$

The holographic force, by contrast, is a stress-tensor effect with $\Delta_T = 2$, which would register as

$$\kappa_T \;=\; 2\Delta_T \;=\; 4. \tag{6}$$

Equations (5) and (6) are the pre-registered dichotomy: a distance reconstructed from the spin–spin correlator must report $\kappa \approx 0.25$, not $\kappa \approx 4$. If the Synge distance secretly carried the force, it would have to read out $\kappa_T = 4$; if it is force-blind, it reads out $\kappa_\sigma = 0.25$.

A wording correction. It is worth being precise about the *shape* of $d(r)$, because an earlier informal statement of this result called it "linear."[2] Explicitly,

$$C(r) = r^{-1/4} \implies d(r) = \epsilon \sqrt{\tfrac{1}{2} \ln r}. \tag{7}$$

The scale ϵ enters Eq. (2) as a pure multiplicative prefactor, so the monotonicity verdict and the log-slope κ are both ϵ-independent by construction; we set $\epsilon = 1$ throughout.

3 Numerical test: a three-tier pre-registered protocol

The test (recorded as run `SIM-2026-031`) was pre-registered with two refutation gates and three tiers. The *geometric* half is refuted if $d(r)$ is non-monotone on the artifact-free synthetic tier; the *force-blindness* half is refuted if the measured κ lands closer to 4 than to 0.25; the test is inconclusive if no fit reaches $R^2 > 0.9$. We took $r = 1, \ldots, 8$ $(= N/2)$ and $\epsilon = 1$.

Tier 1 — synthetic CFT (primary). The clean correlator $C(r) = r^{-1/4}$, with $d(r)$ from Eq. (2). The map returns a monotone $d(r)$ and a log-slope

$$\kappa^{(\mathrm{syn})} = 0.250 \quad \text{(exactly)}, \tag{8}$$

matching Eq. (5). As anticipated in Eq. (7), the square-root-log shape gives only $R^2 = 0.73$ for a *linear* fit of d versus r, which is the basis for the wording correction above.

Tier 2 — exact 16-qubit critical TFIM (physical). We diagonalize the critical transverse-field Ising Hamiltonian $H = -\sum_i X_i - \sum_i Z_i Z_{i+1}$ on $N = 16$ sites with periodic boundary conditions (ground-state energy $E_0 = -20.4046$; $\langle Z_i \rangle$ vanishes to $\sim 10^{-13}$ as the $\mathbb{Z}_2$ symmetry requires), and form the connected correlator $C(r) = \langle Z_i Z_{i+r} \rangle - \langle Z_i \rangle \langle Z_{i+r} \rangle$ averaged over i. The Synge distance is again monotone increasing. To deconvolve finite-size curvature we read the log-slope against the periodic chord variable $(N/\pi)\sin(\pi r/N)$ rather than the raw r, giving

$$\kappa^{(16\mathrm{q})}_{\mathrm{chord}} = 0.243, \tag{9}$$

in clear agreement with $\kappa_\sigma = 0.25$ and far from $\kappa_T = 4$. (The uncorrected raw-r slope is 0.196; the chord correction restores it toward the CFT value, as expected.) The best-fit form for $d(r)$ on this tier is a power law with $R^2 = 0.95$, satisfying the conclusiveness gate.

Tier 3 — pathological control. As a negative control we feed the flat, $r = 4$-anomalous $\langle ZZ \rangle$ proxy data from an earlier record (`SIM-2026-011`) through the same map. As designed, this tier *fails* the monotonicity gate ($d(r)$ non-monotone) and returns a meaningless slope ($\kappa = 0.089$), confirming that the protocol's gates actually discriminate good geometric input from bad rather than passing everything.

[2] We correct that wording here. For the clean CFT correlator $C = r^{-1/4}$, substituting into Eq. (2) gives $d(r) = \sqrt{-2\epsilon^2 \ln r^{-1/4}} = \epsilon\sqrt{\tfrac{1}{2}\ln r}$ — a *square-root-logarithmic* curve. It is monotone increasing but is neither linear in r nor a power law; a linear fit to it returns $R^2 \approx 0.73$ over $r \in [1, 8]$, confirming it is only approximately straight. The pre-registered refutation gates (monotonicity and the κ dichotomy) are unaffected by this shape correction, since both are read from $\ln|C|$ versus $\ln r$, not from the linearity of $d(r)$.

Verdict. Both pre-registered gates point the same way: the distance is monotone (geometric half *not* refuted) and its slope is $\kappa \approx 0.25$, not ≈ 4 (force-blindness half *not* refuted), with the conclusiveness threshold met at $R^2 = 0.95$. The hypothesis is **SUPPORTED**: the Synge distance is a genuine emergent-geometry object that is blind to the holographic force. Table 1 collects the numbers.

Tier	$C(r)$ source	monotone $d(r)$?	κ
1 (synthetic)	$C = r^{-1/4}$	yes	0.250 (exact)
2 (physical)	exact 16q connected $\langle ZZ \rangle$	yes	0.243 (chord)
3 (control)	`SIM-2026-011` proxy	no (as predicted)	0.089 (garbage)
prediction $\kappa_\sigma = 2\Delta_\sigma$		—	0.250
force prediction $\kappa_T = 2\Delta_T$		—	4

Table 1: Three-tier pre-registered test of the Synge map. The measured slopes track the spin-field value $2\Delta_\sigma = 0.25$, not the stress-tensor value $2\Delta_T = 4$: the emergent distance is force-blind.

4 Discussion: why the CMI and Crofton routes failed

The value of this measurement is not the distance itself — which is unsurprising once one writes Eq. (4) — but what it explains in retrospect. Two earlier attempts in this program tried to recover the $\alpha \approx 4$ force from boundary correlation data, and both failed. The present result shows that they failed for the *same structural reason*, not for separate technical ones.

The conditional-mutual-information route. An early hypothesis posited that the holographic force could be read directly from a correlation measure — a conditional mutual information (CMI) between boundary regions. That route did not reproduce $\alpha \approx 4$. The operator-dimension argument says why it could not: any quantity built from the spin–spin correlator is dominated by the σ primary, $\Delta_\sigma = 1/8$, and so reports the physics of $\kappa = 2\Delta_\sigma = 0.25$. The force lives in a different sector entirely, the stress tensor $\Delta_T = 2$. No amount of post-processing of a σ-dominated correlator manufactures a T-dominated exponent.

The Crofton / integral-geometry route. A second family of hypotheses tried to obtain the force from integral geometry — a Crofton or kinematic-space construction that turns correlator-derived distances into a measure on geodesics. These too were refuted. Again the reason is now visible upstream of the integral-geometry machinery: the input distance is the same force-blind, σ-governed object tested here. Integrating a force-blind distance over kinematic space cannot conjure a stress-tensor force; the failure was inherited from the input, not introduced by the transform.

The unifying statement. Distance and force in this system are carried by operators of *different* scaling dimension — $\Delta_\sigma = 1/8$ for the correlator-built geometry, $\Delta_T = 2$ for the force. A geometry reconstructed from a single two-point correlator is therefore force-blind by construction, and no map that takes only that correlator as input ($d = f(C)$, Crofton of d, CMI of the same state's correlations) can recover the stress-tensor force, even in principle. This is a genuine no-go for the "correlator → geometry → force" strategy as a whole, and it is consistent with a companion finding in this program that no state-only normalization of a correlator yields a chain-independent transport degradation: in both cases the obstruction is that a single low-dimension observable does not contain the higher-dimension data one is trying to extract. Recovering the force requires an

observable that couples to T directly — the stress-tensor correlator itself, as used in the $\alpha \approx 4$ measurements — rather than any reprocessing of the σ sector.

5 Conclusion

The Synge world-function map turns a critical-Ising correlator into a legitimate, monotone, emergent distance — but that distance is force-blind. Its log-slope is fixed by the lightest primary in the correlator ($\kappa = 2\Delta_\sigma = 0.25$, measured as 0.243 on the exact 16-qubit ground state) and not by the stress tensor that carries the $\alpha \approx 4$ force ($\kappa_T = 2\Delta_T = 4$). Because distance and force are carried by operators of different dimension, a geometry reconstructed from one two-point correlator cannot reproduce the holographic force, which explains in a single clean measurement why the CMI and Crofton routes failed for structurally the same reason. The constructive lesson is equally clean: to see the force, measure the operator that carries it.

References

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